Clinical Risk Prediction of Diabetes Using Supervised Machine Learning Models on Patient Health Records

Final Project Report | Deep Learning Project (ICT-4442) | Team No. 11

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Abstract

This project compares Logistic Regression, Random Forest, Support Vector Machine (RBF kernel), and a Multi-Layer Perceptron (MLP) neural network for binary diabetes classification using the Pima Indians Diabetes dataset. The supplied dataset contains 768 records and 8 clinical input features; 500 records have outcome 0 and 268 have outcome 1. Preprocessing is performed inside model pipelines to reduce leakage: biologically implausible zero values in selected measurements are treated as missing, median imputation is fitted on training data, and numeric features are standardised. Models are evaluated on a held-out stratified test set using Accuracy, Precision, Recall, F1-score, and ROC-AUC, alongside stratified five-fold cross-validation on the training data. Results are intended for educational comparison and not clinical diagnosis.

1. Objectives

1. Compare classical supervised models and an MLP neural network using consistent data splits and clinically relevant classification metrics.
2. Identify features with high predictive importance in the Random Forest model and discuss model limitations.

2. Dataset and preprocessing

The CSV has columns Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, and Outcome. Outcome is the binary target (0 = non-diabetic, 1 = diabetic). The split is 80% training and 20% testing with stratification and random_state=42. Zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI are converted to missing values, then median-imputed. StandardScaler is included within each pipeline. Cross-validation is conducted on the training partition only.

3. Models

• Logistic Regression (C=1.0, max_iter=1000) — linear baseline.
• Random Forest (100 trees, max_depth=10, random_state=42) — tree ensemble with feature importance.
• SVM (RBF kernel, C=1.0, gamma='scale') — nonlinear kernel classifier.
• MLP Neural Network (64 and 32 hidden units, ReLU activation, early stopping) — feed-forward neural network baseline.

4. Held-out test results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.747	0.660	0.574	0.614	0.810
SVM (RBF)	0.740	0.652	0.556	0.600	0.796
MLP Neural Network	0.721	0.617	0.537	0.574	0.818
Logistic Regression	0.708	0.600	0.500	0.545	0.813

On this particular held-out split, Random Forest achieved the highest F1-score (0.614) and accuracy (0.747) among the four models. The MLP had ROC-AUC 0.818. Results differ by metric, so no model should be described as universally best. Recall is especially important in screening settings because false negatives are missed positive cases.

5. Stratified five-fold cross-validation on training data

Model	Accuracy mean	Precision mean	Recall mean	F1 mean	AUC mean
Logistic Regression	0.788	0.764	0.575	0.655	0.843
SVM (RBF)	0.780	0.742	0.584	0.648	0.833
Random Forest	0.769	0.698	0.603	0.645	0.824
MLP Neural Network	0.770	0.716	0.589	0.639	0.848

Cross-validation ranked Logistic Regression highest by mean F1-score (0.655). This differs from the held-out split ranking, illustrating the variability expected from a small dataset. Use both evaluations when interpreting model performance.

6. Random Forest feature importance

The five highest Random Forest feature importance values were: Glucose (29.2%); BMI (16.1%); DiabetesPedigreeFunction (12.3%); Age (11.4%); Insulin (9.5%). Feature importance is model-specific and indicates predictive contribution in this fitted model; it does not prove causation.

7. Limitations and conclusion

The dataset is relatively small and represents a specific population; results may not generalise to other populations. The test set contains limited examples, so metrics are uncertain. Zero-value handling follows a common convention for this dataset but may not reflect the original clinical record. The models are educational prototypes and must not be used to diagnose or make treatment decisions. Future work could assess threshold selection, calibration, confidence intervals, fairness, and external validation on independent datasets.

References

1. Smith, J. W. et al. (1988). “Using the ADAP learning algorithm to forecast the onset of diabetes mellitus.” Proceedings of the Symposium on Computer Applications in Medical Care.
2. Breiman, L. (2001). “Random Forests.” Machine Learning, 45, 5–32.
3. Pima Indians Diabetes dataset, commonly distributed through UCI Machine Learning Repository and Kaggle. Verify the exact repository/version used before formal citation.